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**CSA1021 INTRODUCTION TO PROBLEM SOLVING
AND PROGRAMMING**

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BRANCH :CSE AI ML

SEMESTER : Fall Semester 2026-27

A Command-Line Hospital Information and Patient Management System

1. INTRODUCTION

The **Sreeam Hospital Portal** is a Python-based command-line application developed to demonstrate the basic functioning of a hospital information management system. The project provides a simple and structured interface through which users can access information related to admitted patients, assigned doctors, consultation status, and new patient allotment.

The application is designed around a **menu-driven approach**, making it easy for users to navigate between different hospital-related operations. When the program starts, it displays a welcome message followed by five major options: viewing admitted patients, viewing doctors and their departments, checking consultation status, allotting a new patient, and exiting the portal.

The project demonstrates several fundamental concepts of Python programming, including:

- Variables and data types
- Lists
- User input and output
- if-elif-else decision-making
- while loops
- for loops
- Nested conditional statements
- String formatting
- Formatted output
- Program termination using break

The project is primarily intended as an **academic demonstration of problem-solving and programming concepts**. It represents how a real-world hospital scenario can be converted into a structured computational solution.

2. PROBLEM STATEMENT

Managing hospital information manually can become difficult when patient, doctor, and consultation details increase. Even in a small hospital environment, users may need to quickly access information such as:

- Which patients are currently admitted?
- Which doctor is assigned to a particular patient?
- What department does a doctor belong to?

- What is the consultation status of a patient?
- How can information about a newly admitted patient be entered?

The objective of this project is to develop a **simple command-line hospital portal** that organizes these operations into a single interactive system.

The proposed system provides a centralized menu through which the user can select the required operation. Based on the selected option, the program displays the appropriate information or accepts new patient details.

The project therefore converts a real-world hospital information problem into a structured programming problem using Python.

3. OBJECTIVES OF THE PROJECT

The major objectives of the Sreeam Hospital Portal are:

3.1 Primary Objectives:

1. To develop a menu-driven hospital information system using Python.
2. To provide quick access to predefined patient information.
3. To display doctor and department information.
4. To provide consultation-status information.
5. To allow entry of new patient information.
6. To provide a simple and understandable command-line interface.
7. To demonstrate fundamental Python programming concepts.

3.2 Learning Objectives

Through this project, the developer gains practical understanding of:

- Problem decomposition
- Algorithm development
- Conditional statements
- Iterative programming
- List manipulation
- User-input processing
- Console-based application development
- Testing and debugging
- Basic software documentation

4. FUNCTIONAL REQUIREMENTS

Functional requirements describe what the system is expected to do.

4.1 Main Menu

When the application starts, the system displays a menu containing the following operations:

1. Admitted Patients
2. Doctors / Departments Assigned
3. Consultation Status
4. New Patient Allotment
5. Exit

The user selects an operation by entering the corresponding number.

4.2 Admitted Patient Module

The admitted-patient module displays the list of currently available patient records.

For a selected patient, the system displays information such as:

- Patient name
- Patient ID
- Age
- Gender
- Room number
- Diagnosis
- Assigned doctor
- Contact details

4.3 Doctor Module

The doctor module provides information about doctors available in different departments.

The system can display:

- Doctor name
- Doctor ID
- Age
- Gender
- Room number

- Qualifications
- Assigned department
- Specialization
- Contact details

4.4 Consultation Module

The consultation module provides information about selected patients and their consultation status.

The system can display whether the consultation is:

- Completed
- Pending

It can also display information regarding medicines purchased.

4.5 New Patient Allotment

The system allows the user to enter:

- Patient name
- Patient age
- Diagnosis
- Assigned doctor

After receiving the information, the program displays an admission confirmation.

4.6 Exit Module

The user can terminate the portal by selecting option 5.

The system displays an exit confirmation message before terminating execution.

5. NON-FUNCTIONAL REQUIREMENTS

Non-functional requirements describe the qualities and operational characteristics of the system.

5.1 Usability

The system uses a numbered menu so that users can understand and operate it without complicated commands.

5.2 Simplicity

The application uses basic Python constructs, making it suitable for beginners and easy to understand.

5.3 Portability

Since the project is implemented in Python and uses standard language features, it can be executed in a Python-supported command-line environment.

5.4 Performance

The current application works with a small amount of data, so its operations are lightweight and execute quickly.

5.5 Maintainability

The program structure is straightforward, allowing individual modules to be modified or extended in future versions.

5.6 Reliability

For valid inputs, the program follows predefined branches and produces the corresponding output.

6. SYSTEM ARCHITECTURE

The Sreeam Hospital Portal follows a **menu-driven procedural architecture**.

The basic flow of the application can be represented as:

User → Main Menu → Selected Module → Processing → Output → Main Menu

The application continuously waits for user input. Depending on the selected option, control is transferred to the appropriate section of the program.

Major Components

1. User Interface Layer
Responsible for displaying menus and accepting input.
2. Patient Management Component
Handles admitted patient information.
3. Doctor Management Component
Handles doctor and department information.
4. Consultation Component
Displays consultation-related information.
5. Patient Admission Component
Accepts information about a new patient.
6. Control Flow Component
Controls program execution using loops and conditional statements.
7. Exit Component
Terminates the application when requested.

7. DESIGN DIAGRAMS

7.1 Use Case Diagram

The primary actor of the current system is the **User**.

The user can perform the following operations:

- View admitted patients
- View doctor information
- Check consultation status
- Allot a new patient
- Exit the portal

Use Case Description

Actor	Use Case	Description
User	View Patients	Displays admitted patient information
User	View Doctors	Displays doctor and department details

User	Check Consultation	Displays consultation status
User	New Patient Allotment	Accepts new patient information
User	Exit	Terminates the application

8. WORKFLOW OF THE SYSTEM

The workflow begins when the program is executed.

Step 1

The system displays the welcome message.

Step 2

The main menu is displayed.

Step 3

The user selects an option.

Step 4

The system validates the selected menu branch.

Step 5

If the selection requires additional information, the user is asked to provide it.

Step 6

The system processes the selection.

Step 7

The appropriate result is displayed.

Step 8

The system returns to the main menu.

Step 9

The process continues until the user selects **Exit**.

9. DESIGN DECISIONS AND RATIONALE

Several design decisions were made while developing the project.

Menu-Driven Interface

A menu-driven interface was selected because the project contains multiple independent operations. A numbered menu makes these operations easy to access.

Lists for Records

Lists were used to store predefined names and identifiers because lists are simple and appropriate for demonstrating introductory Python data structures.

Conditional Branching

if, elif, and else statements were used to determine which operation should be performed based on user input.

Loop-Based Execution

The main program uses a continuous loop so that the user can perform multiple operations without restarting the application.

Console-Based Design

A command-line interface was selected because it keeps the project lightweight and demonstrates the underlying programming logic without depending on a graphical environment.

10. IMPLEMENTATION DETAILS

The project is implemented using Python.

The major programming constructs used are:

`input()`

Used to accept information from the user.

`print()`

Used to display menus, records, messages, and results.

Lists

Lists are used to store predefined patient, doctor, and consultation identifiers.

if-elif-else

These statements determine which operation should be executed.

for Loop

The for loop is used to display items stored in lists.

while Loop

The main menu operates repeatedly through a loop.

break

The break statement is used to terminate the loop when the user chooses to exit or when certain invalid selections are encountered.

11. PATIENT MANAGEMENT MODULE

The patient-management module provides access to admitted-patient records.

The current implementation contains predefined patient records including:

- Manish
- Rishabh
- Geeta
- Lokesh

After the user selects a patient, the system displays the corresponding information.

Example

Patient: Manish

Patient ID: P001

Age: 20

Gender: Male

Room: AB101

Diagnosis: Bone Fracture

Assigned Doctor: Dr. Goswami

This module demonstrates how a user selection can be mapped to a specific data record.

12. DOCTOR MANAGEMENT MODULE

The doctor-management module provides information about doctors and their departments.

The current records include:

- Dr. Goswami
- Dr. Rao
- Dr. Singh
- Dr. Saxena

For each doctor, the system provides information such as qualifications, department, specialization and contact details.

This module demonstrates the use of nested decision-making and structured console output.

13. CONSULTATION MANAGEMENT MODULE

The consultation module allows the user to select a patient ID and view the corresponding consultation information.

The current system demonstrates different consultation states, including:

- Completed consultation
- Pending consultation

The module also displays whether medicines have been purchased.

This demonstrates how status information can be associated with a particular patient identifier.

14. NEW PATIENT ALLOTMENT MODULE

The new-patient module provides an input-based workflow.

The user is asked to enter:

1. Patient name
2. Patient age
3. Diagnosis
4. Assigned doctor

The program then displays a confirmation message indicating that the patient has been admitted under the selected doctor.

This part of the application is particularly important because it demonstrates the transition from **predefined data to user-generated input**.

15. USER INTERFACE

The project uses a simple terminal interface.

The interface follows the principle:

Simple Input → Clear Processing → Immediate Output

This approach makes the application easy to operate and allows the user to understand the result of every action.

Interface Characteristics

- Numbered menu
- Clear prompts
- Simple terminology
- Immediate feedback
- Repeated menu access
- Exit confirmation

16. RESULTS

Sreeam Hospital Portal — Terminal Output

Admitted Patient Record - Manish

```
welcome to Sreeam hospital portal
1. Admitted Patients
2. Doctor(department) assigned
3. Consultation status
4. New Patient Allotment
5. To Exit
what u wish to do today? Admitted Patients:
1. Manish
2. Rishabh
3. Geeta
4. Lokesh
Select Patient: Selected Patient : 1. Manish
Patient ID : P001
Age : 20
Gender : Male
Room no : AB101
Diagnosed with : Bone Fracture
Assigned Doctor : Dr. Goswami
Contact Details : 98700XXXXX
what u wish to do today? Portal Exit Successful!!
Thank You
```

Sreeam Hospital Portal — Terminal Output

Doctor / Department Details - Dr. Singh

```
welcome to Sreeam hospital portal
1. Admitted Patients
2. Doctor(department) assigned
3. Consultation status
4. New Patient Allotment
5. To Exit
what u wish to do today? Doctors(departments) Assigned:
1. Dr. Goswami
2. Dr. Rao
3. Dr. Singh
4. Dr. Saxena
Select Doctor: Selected Doctor : 3. Dr. Singh
Doctor ID : D003
Age : 41
Gender : Male
Room no : CD103
Qualifications : MBBS, MD, DNB
Assigned Department : Cardiology
Specialisation in : Heart Failure & Transplantation
Contact Details : 72907XXXXX
what u wish to do today? Portal Exit Successful!!
Thank You
```

Sreeam Hospital Portal — Terminal Output

Consultation Status - P034

```
welcome to Sreeam hospital portal
1. Admitted Patients
2. Doctor(department) assigned
3. Consultation status
4. New Patient Allotment
5. To Exit
what u wish to do today? Patient ID of Patients for consultation:
1. P026
2. P034
3. P027
Select Patient: Selected Patient: 2. P034
Consultation Status: Pending
Consultation time: 10:40 am
Medicines purchased: Nil
what u wish to do today? Portal Exit Successful!!
Thank You
```

Sreeam Hospital Portal — Terminal Output

New Patient Allotment

```
welcome to Sreeam hospital portal
1. Admitted Patients
2. Doctor(department) assigned
3. Consultation status
4. New Patient Allotment
5. To Exit
what u wish to do today? Enter Name of Patient: Enter Patient Age: Patient Diagnosed with: Doctor
assigned: Patient Ananya under Doctor Dr. Goswami has been admitted successfully
Patient ID and Room No will be generated shortly!!
what u wish to do today? Portal Exit Successful!!
Thank You
```

Sreeam Hospital Portal — Terminal Output

Portal Exit

```
welcome to Sreeam hospital portal
1. Admitted Patients
2. Doctor(department) assigned
3. Consultation status
4. New Patient Allotment
5. To Exit
what u wish to do today? Portal Exit Successful!!
Thank You
```

Sreeam Hospital Portal — Terminal Output

Invalid Patient Selection

```
welcome to Sreeam hospital portal
1. Admitted Patients
2. Doctor(department) assigned
3. Consultation status
4. New Patient Allotment
5. To Exit
what u wish to do today? Admitted Patients:
1. Manish
2. Rishabh
3. Geeta
4. Lokesh
Select Patient: Patient not found!!
```

17. TESTING APPROACH

Testing was performed to determine whether each major operation produces the expected result.

The testing process covers:

- Main menu testing
- Valid patient selection
- Invalid patient selection
- Valid doctor selection
- Consultation-status selection
- New patient entry
- Exit operation

Testing helps identify logical errors and ensures that the program behaves as intended for the tested inputs.

18. TEST CASES

Test Case	Input	Expected Result
TC01	1	Patient menu displayed
TC02	1 → 1	Manish's details displayed
TC03	1 → Invalid number	Patient not found message
TC04	2 → 3	Dr. Singh's details displayed
TC05	3 → 2	P034 consultation displayed
TC06	4 + patient details	Admission confirmation displayed
TC07	5	Portal exits successfully

19. CHALLENGES FACED

During the development of the project, several programming challenges were considered.

19.1 Managing Multiple Menu Operations

The program contains multiple independent operations. Maintaining clear control flow between them requires careful use of conditional statements.

19.2 Handling User Input

The application accepts integer choices as well as textual information. Correctly handling these different types of input is important.

19.3 Repetitive Data

The predefined patient and doctor records contain repeated output structures. A more advanced implementation could reduce repetition using dictionaries, functions or classes.

19.4 Input Validation

The current implementation expects certain inputs to be numeric. Invalid non-numeric input can require additional exception handling.

19.5 Data Persistence

The current project does not permanently store newly entered patient information. Implementing a database would solve this limitation.

20. LIMITATIONS

Although the project successfully demonstrates its intended programming concepts, it has certain limitations.

1. Data is primarily predefined.
2. No database is currently connected.
3. There is no login or authentication system.
4. Patient information is not permanently stored.
5. There is no search facility.
6. There is no update or delete operation.
7. There is no graphical user interface.
8. Advanced input validation is limited.
9. Hospital billing is not included.
10. Real-time appointment management is not implemented.

These limitations provide opportunities for future development.

21. Security, Privacy and Data Considerations

It covers how the hospital portal should handle sensitive patient information.

For your **Sreeam Hospital Portal**, it includes:

- Patient information should be kept confidential.
- A real hospital system should have **login/authentication**.
- Different users should have different permissions, such as **admin, doctor, receptionist**.
- Patient records should be stored securely.
- The system should maintain **access logs** to track who viewed or changed information.
- Input validation should prevent incorrect or harmful data.
- Real patient information should not be used in an academic project; **fictional/anonymized data** should be preferred.
- Future versions could add encryption and secure database storage.

22. CONCLUSION

The **Sreeam Hospital Portal** successfully demonstrates the development of a menu-driven hospital information application using Python.

The project transforms a real-world hospital scenario into a structured programming solution. Through its patient, doctor, consultation and new-patient modules, the application demonstrates how fundamental programming concepts can be combined to create a functional application.

The project also provides valuable practical experience in:

- Problem solving
- Program design
- Python syntax
- Control structures
- Data organization
- User interaction
- Testing
- Documentation

Although the current implementation is intentionally simple and educational, it establishes a strong foundation for developing a more advanced hospital management system.

With database integration, authentication, improved validation, graphical interfaces and additional hospital modules, the project could be extended into a much more comprehensive application.

Overall, the Sreeam Hospital Portal represents a practical application of Python programming concepts to a real-world information-management scenario.